

Problem 49. Find all distinct prime numbers p, q and r such that

$$3p^4 - 5q^4 - 4r^2 = 26.$$

Problem 50. Let ABC be a triangle. The internal bisector of $\angle ABC$ intersects the side AC at L and the circumcircle of triangle ABC again at $W \neq B$. Let K be the perpendicular projection of L onto AW . The circumcircle of triangle BLC intersects line CK again at $P \neq C$. Lines BP and AW meet at point T . Prove that $AW = WT$.

Problem 51. Let $x_0 < x_1 < x_2 < \dots$ be an infinite sequence of positive integers. Prove that there exists a unique integer $n \geq 1$ such that

$$x_n < \frac{x_0 + x_1 + \dots + x_n}{n} \leq x_{n+1}.$$

Problem 52. Find the smallest real constant C such that for any positive real numbers a_1, a_2, a_3, a_4 and a_5 (not necessarily distinct), one can always choose distinct subscripts i, j, k and ℓ such that

$$\left| \frac{a_i}{a_j} - \frac{a_k}{a_\ell} \right| \leq C.$$